Yuwen Tan

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EDUCATIONAL BACKGROUND

Boston University *Boston, MA*
Ph.D., Department of Computer Science — Advisor: Prof. Boqing Gong — 09/2024–Present

Huazhong University of Science and Technology *Wuhan, China*
M.S., Department of Artificial Intelligence and Automation — Advisor: Prof. Xiang Xiang — 09/2021–06/2024

Huazhong University of Science and Technology *Wuhan, China*
B.E., Department of Artificial Intelligence and Automation — 09/2017–06/2021

PUBLICATIONS

- **Tan Y**, Huang J, Huang J, Li H, Gong B. Natural Language Camera Movement Understanding. European Conference on Computer Vision (ECCV 2026)
- **Tan Y**, Qing Y, Gong B. Vision LLMs Are Bad at Hierarchical Visual Understanding, and LLMs Are the Bottleneck. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)
- **Tan Y**, Gong B. Lifting Data-Tracing Machine Unlearning to Knowledge-Tracing for Foundation Models. Transactions on Machine Learning Research (TMLR 2026)
- **Tan Y**, Zhou Q, et al. Semantically-Shifted Incremental Adapter-Tuning is A Continual ViTransformer. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)
- **Tan Y**, Xiang X. Cross-Domain Few-Shot Incremental Learning for Point-Cloud Recognition. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2024)
- **Tan Y**, Xiang X, et al. Coupling Bracket Segmentation and Tooth Surface Reconstruction on 3D Dental Models. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2023)
- **Tan Y**, Xiang X, et al. Boundary-Constrained Graph Network for Tooth Segmentation on 3D Dental Surfaces. The 14th International Workshop on Machine Learning in Medical Imaging (MLMI 2023)
- Xiang X, **Tan Y**, Wan Q, et al. Coarse-to-fine incremental few-shot learning. European Conference on Computer Vision (ECCV 2022)
- Xiang X, Wang F, **Tan Y**, et al. Imbalanced regression for intensity series of pain expression from videos by regularizing spatial-temporal face nets. Pattern Recognition Letters (PRL 2022)

RESEARCH EXPERIENCE

Machine Unlearning in Foundation Models — 09/2024–02/2025
Vision and Graphics Group, BU — Instructor: Prof. Boqing Gong

- Designed realistic unlearning settings for vision-language foundation models, constructed a comprehensive unlearning benchmark, and refined the evaluation metrics to further advance research on unlearning in foundation models.
- Developed novel machine unlearning algorithms to achieve effective and scalable unlearning while preserving model quality. Investigated different strategies to ensure the balance between unlearning effectiveness, computational efficiency, and real-world applicability.

Object Recognition in Open Environments — 03/2023–06/2024
HUST AIA Image & Vision Learning Lab, HUST — Instructor: Prof. Xiang Xiang

- Constructed training sets with the known categories and defined the unknown categories as unknown sub-classes under the same coarse granularity. Labeled the attributes of the dataset and built attribute classifiers.
- Identified the unknown targets based on attribute knowledge graph and category hierarchy.

- Designed incremental algorithms to enable the model to continuously learn the fine-grained unknown classes.

Orthodontic Assistance System Based on 3D Dental Models 03/2022–03/2023
HUST AIA Image & Vision Learning Lab & West China Hospital, Sichuan University Instructor: Prof. Xiang Xiang

- Collected and labeled 3D dental models to build the tooth segmentation and bracket segmentation dataset.
- Proposed the robust tooth segmentation method which could process 3D dental models with different morphologies. The algorithm aimed at improving the performance of boundary segmentation through several auxiliary losses.
- Proposed the graph-based bracket segmentation network with residual connection and dilated K-NN construction which could segment brackets on 3D dental models automatically and then designed a hole-filling method to construct the 3D dental models with brackets removed.

WORK EXPERIENCE

Google (YouTube) 05/2026–Present
Student Researcher

- Research Topic: Generative Video Evaluation
- Supervisor: Dr. Yilin Wang

Pixocial, Human-First AI 05/2025–08/2025
Applied Research Intern

- Research Topic: Video Generation Acceleration and Evaluation
- Supervisor: Dr. Haoxiang Li

TEACHING ASSISTANT

Computer Vision (Spring 2023) 02/2023–06/2023

- Instructor: Prof. Xiang Xiang (School of Artificial Intelligence and Automation)
- 2 credits; Undergraduate course

Pattern Recognition (Spring 2023) 02/2023–06/2023

- Instructor: Prof. Xiang Xiang (School of Artificial Intelligence and Automation)
- 3 credits; Graduate course

HONORS & AWARDS

- Hariri Institute 2025 Graduate Student Fellow Award, Boston University 05/2025
- Excellent Graduate, Huazhong University of Science and Technology 06/2024
- Merit Student, Huazhong University of Science and Technology 09/2022
- First Prize Scholarship, Huazhong University of Science and Technology 09/2021
- Social Responsibility Award, Huazhong University of Science and Technology 09/2019
- Excellent New Student Award, Huazhong University of Science and Technology 02/2018

PROFESSIONAL SKILLS

Communication: Mandarin (Native), English.
Coding Languages: Python (PyTorch, TensorFlow), MATLAB, C

SERVICE

- **Conference Reviewer:** IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); European Conference on Computer Vision (ECCV); Conference on Neural Information Processing Systems (NeurIPS); International Conference on Learning Representations (ICLR).
- **Journal Reviewer:** IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI); International Journal of Computer Vision (IJCV); ACM Computing Surveys (CSUR).